

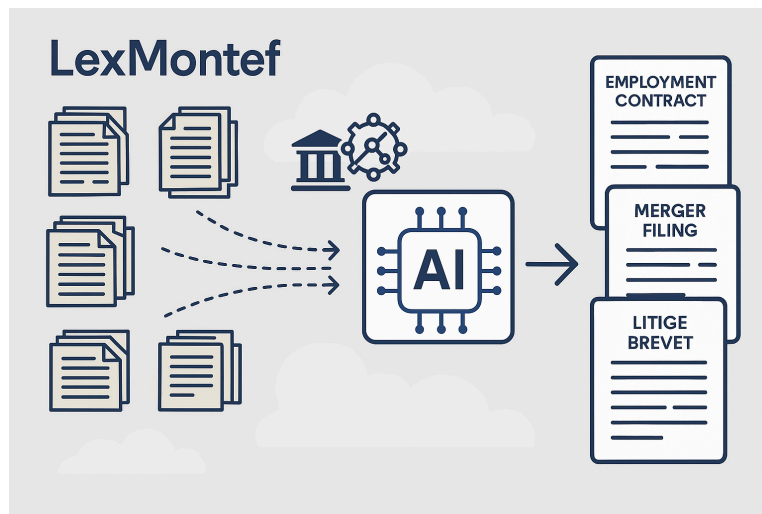
Exam - 2025

INFO9023: Machine Learning Systems Design

Instructions

- Oral exam
- You will receive **1 use case** and a series of questions related to it
- Make sure to **thoroughly read** the use case description and each question
- **Motivate** your answers. Often the reason for making a specific design choice is as important as the choice itself.
- The total grade for the exam is **20 points**. Points are mentioned for each question

Use case - Classifying legal documents for archiving



LexMontef, a legal technology company, is building an ML-powered search and categorization engine for law firms. Over the past two decades, large volumes of legal documents - including contracts, court decisions, case summaries, and regulatory filings - have been scanned and digitized by law firms. However, a large part of these documents remain unstructured, making legal research and knowledge reuse inefficient.

LexMontef wants to create an ML system that automatically classifies documents into predefined categories (e.g. “employment contract,” “merger filing,” “patent dispute,” etc.). The documents will be in English and French.

The output of the ML model classifying the documents might be consumed in different forms by other downstream systems.

LexMontef is setting up a team for building this ML application and you have been appointed as Tech-Lead. LexMontef is aware of the importance of building a robust and scalable system. You are tasked with taking key design decisions regarding this application.

Question 1 - Data preparation

1.1. What **type** of data sources might be used for training this model? *[Refer to the dimensions covered during the class]*
(2 points)

Question 2 - Model training & optimisation

2.1. You want to optimize the training of your model by parallelising it. Name and briefly explain the three types of model training parallelisation covered in this course..

(3 points)

Question 3 - Model serving

3.1. Will you implement batch or real-time serving of your ML model? Will it be synchronous or asynchronous? Motivate your answer.

(2 point)

3.2. Will you deploy your model on premise or in the cloud? Motivate your answer. Also give an example of a use case where the other deployment method would be applied.

(2 point)

3.3. Explain the concept of latency, throughput, auto-scaling and load balancing? How do they relate to each other, will some impact others?

(2 point)

3.4. A downstream system needs to be fed the predictions of your AI model. You need to build an API to access documents. Some of your team members hesitate building a REST or an RPC API. Explain the difference and which one is most appropriate in this context?

(2 point)

Question 4 - Model pipeline

4.1. What are the standard components (steps) of an ML pipeline? Which ones will you focus on in this project?

(3 points)

4.2. What types of triggers exist to automatically launch the ML pipeline? Which one would you recommend using in this use case?

(1 points)

Question 5 - Monitoring

5.1. What kind of drift might occur over time in your system? Give an example in this use case for each type of drift.

(3 point)